RAW coreless filament winding (beam): today vs. 2050
beam span = 6 m, reference service life 50 yr

climate change today
climate change 2050
ozone depletion today
ozone depletion 2050
particulate matter formation today
particulate matter formation 2050
acidification today
acidification 2050
freshwater eutrophication today
freshwater eutrophication 2050
land use today
land use 2050
water use today
water use 2050
mineral resources today
mineral resources 2050

−20
0
20
40
60
80
100
% of largest burden or benefit in category (both backgrounds)

RAW coreless filament winding (beam)
RAW coreless filament winding (beam) - scaled up
flax fibre CFW beam (no in-line sensing)
glass fibre CFW beam
benefits (lighter, left)
net impact